List = [0, 1, 2, 3, 4, 5]

0	1	2	3	4	5

List[0] = 0

List[1] = 1

List[2] = 2

List[3] = 3

List[4] = 4

List[5] = 5

List[0:] = [0,1,2,3,4,5]

List[:] = [0,1,2,3,4,5]

List[2:4] = [2, 3]

List[1:3] = [1, 2]

List[:4] = [0, 1, 2, 3]

List in Python

List = [10 , 'Favtutor', 10, [5,10,15]]

List[0] List[1] List[2] List[3]

- ✓ **Ordered:** Items have defined order which cannot be changed
- ✓ **Mutable:** Items can be modified anytime
- ✓ **Allow duplicates:** Items with the same value is allowed

MyScript Notes

BASICS of List

MUTABLE

LIST

ARRAY

["water", "milk", "ginger", "cardamom"]

0 1 2 3

["water", "milk", "black tea", "ginger", "cardamom"]

0 1 2 3 4

MyScript Notes

```
ingredients = ["water", "milk", "black tea"]

ingredients.append("sugar")
print(f"Ingredients are: {ingredients}")

ingredients.remove("water")
print(f"Ingredients are: {ingredients}")

→ ["water", "milk", "black tea",
   "sugar"]
→ ["milk", "black tea", "sugar"]

spice_option = ["ginger", "cardamom"]
chai_ingredients = ["water", "milk"]
ch                ents.extend(spice_options)
```

MyScript Notes

print(f" chai: {chai_ingredients}")

["water", "milk", "ginger", "cardamom"]

chai_ingredients.insert(2, "black tea")

print( )

["water", "milk", "black tea", "ginger", "cardamom"]

adding with insert at possition number 2 last_added = chai_ingredients.pop()

print(f"{last_added}")

⇓

["water", "milk", "black tea",
"ginger"]

- .reverse() • (max())
- .sort() • (min())

sugar_levels = [1, 2, 3, 4, 5]
print(f" Max sugar level: {max(
(sugar_levels)}

⇓

[

LIST în Python

1. Ce este o List ?

List = colec ie de elemente

Este ordonat

Este MUTABIL (poate fi modificat)

Permite elemente de tipuri diferite

2. Cum se declar ?

l = [1, 2, 3]

Sau cu tipuri diferite:

l = [10, "Ana", 3.14, True]

3. Indexare

10 20 30 40

0 1 2 3

l = [10, 20, 30, 40]

l[0] # 10

l[2] # 30

l[-1] # 40

4. Slicing

l = [1, 2, 3, 4, 5]

l[1:4] # [2, 3, 4]

l[:3] # [1, 2, 3]

l[::2] # [1, 3, 5]

5. Modificare element (diferen fa de tuple)

l = [10, 20, 30]

l[0] = 100

Rezultat: [100, 20, 30]

Lista este mutabil

MyScript Notes

append() – adaug la final

```
l = [1, 2]
l.append(3)
#  [1, 2, 3]
```

insert() – adaug pe pozi ie

```
l.insert(1, 99)
#  [1, 99, 2, 3]
```

remove() – terge valoare

```
l.remove(99)
```

pop() – terge dup index

```
l.pop()    #  terge ultimul
l.pop(0)    #  terge index 0
```

len() – lungime

```
len(l)
```

7. Parcurgere (loop)

```
l = [10, 20, 30]
for x in l:
    print(x)
```

LIST

List Comprehension

```
nums = [1, 2, 3, 4]
patrate = [x*x for x in nums]
#  [1, 4, 9, 16]
```

[ ]

Ordonat

Mutabil

Permite index & slicing

append()

insert()

remove()

MyScript Notes

list comprehension